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The Creating of a Lightweight Intrusion Detection System for MQTT-based IoT Networks

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INTRODUCTION

IoT devices are growing rapidly, but their resource constraints make traditional security tools unsuitable. MQTT is a lightweight messaging protocol widely used in IoT, but it lacks built-in encryption, authentication, and access control, making it vulnerable to exploitation.

This project investigates whether a lightweight, MQTT-aware Intrusion Detection System (IDS) can effectively detect protocol-specific attacks in a resource-constrained environment.

METHODS

Controlled IoT testbed:

- Raspberry Pi 5 (MQTT broker + IDS host)
- ESP8266 (legitimate IoT sensor)
- Kali Linux attacker machine.

Three detection modes tested:

- Signature-based IDS
- Anomaly-based IDS
- Hybrid IDS

Four MQTT attack scenarios:

- Oversized payloads
- Forbidden topic access
- Message flooding
- Reconnaissance/subscription attempts

DISCUSSION

The hybrid IDS provided the broadest attack coverage by combining deterministic signature rules with behavioural anomaly detection.

The Raspberry Pi maintained low CPU usage throughout all tests (scan QR code), confirming suitability for resource-constrained environments.

Limitations include a small single-sensor testbed, a manually defined ruleset, and no encrypted traffic analysis — areas identified for future work.

The IDS was externally reviewed by a cybersecurity professional working as a security analyst and incident responder, who confirmed the architecture and detection logic reflect realistic IoT attack behaviour.

RESULTS

The hybrid IDS achieved full detection coverage across all four attack scenarios, outperforming both signature-based and anomaly-based modes individually.

✓ = Alert generated
✗ = No alert

Attack Type	Signature-Based IDS	Anomaly-Based IDS	Hybrid IDS
Oversized Payload	✓	✓	✓
Forbidden Topic	✓	✗	✓
Flooding	✗	✓	✓
Recon/Sub Attempts	✓	✗	✓

Table 1 - IDS detection performance across three detection modes and four MQTT attack scenarios.

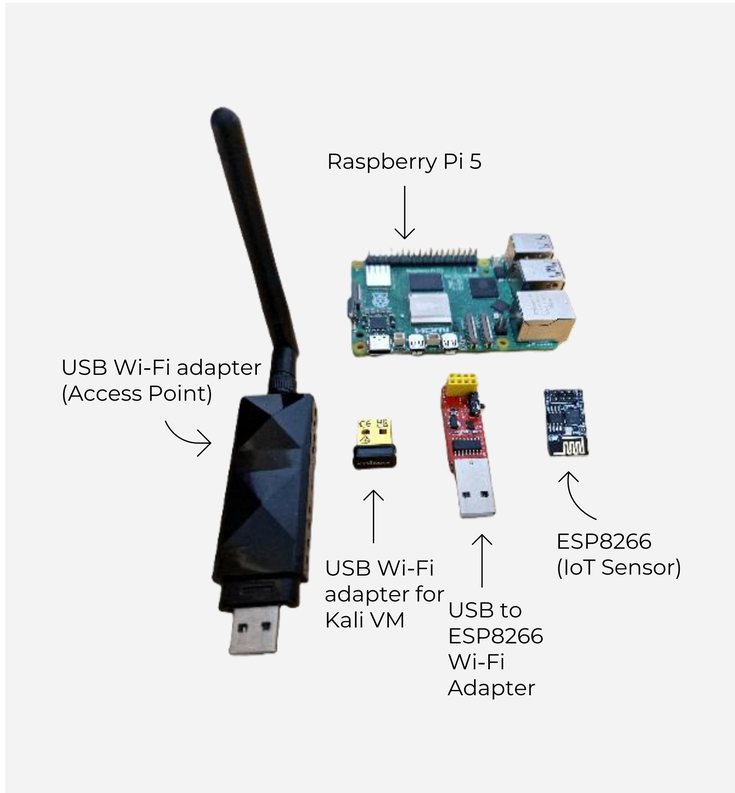

Figure 1 - Physical devices used to build the experimental IoT testbed.

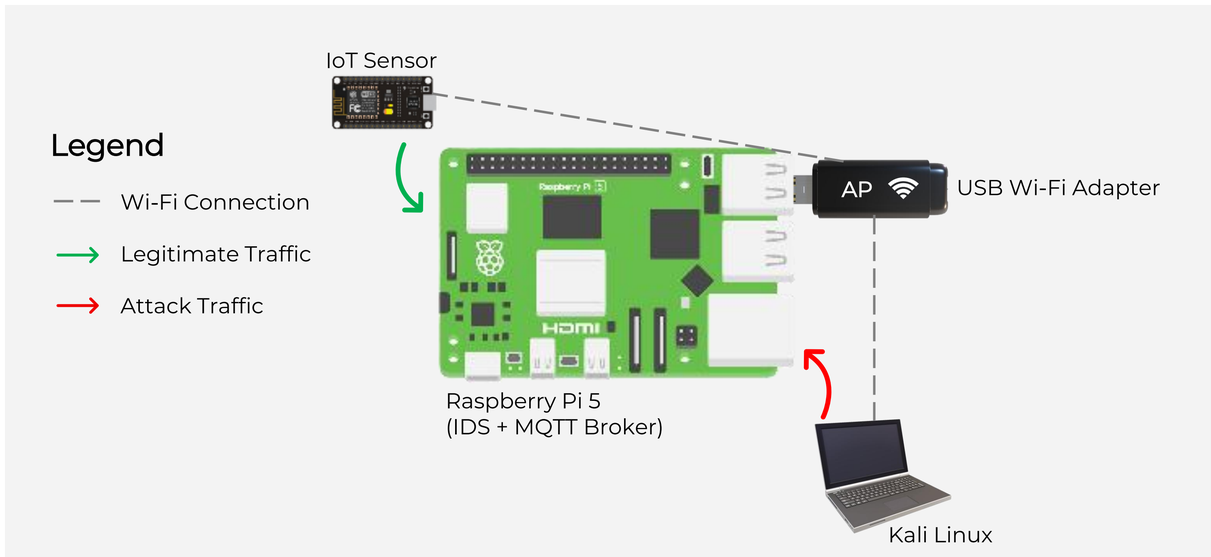

Figure 2 - Network topology of the isolated IoT testbed showing device interactions.

CONCLUSION

The lightweight Python-based IDS successfully detected MQTT-specific attacks across all three detection modes, with hybrid mode achieving full coverage of all four attack types.

This project demonstrates that a protocol-aware, rule-based IDS is feasible for constrained IoT environments without requiring machine learning, contributing a practical proof-of-concept for future IoT security research.

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